2D steady heat-diffusion problem with a sink

Roger Stan Hervet

September 2026

2D Steady Heat-Conduction Problem with a Sink

Consider a 2D rectangular plate of dimensions

$$L_x = 1\,\mathrm{m}, \qquad L_y = 0.5\,\mathrm{m},$$

with constant thermal conductivity

$$k = 1000\,\mathrm{W}/(\mathrm{m}\cdot\mathrm{K}).$$

The plate has a uniform thickness

$$b = 0.01\,\mathrm{m}.$$

The steady-state temperature field is governed by

$$\frac{\partial}{\partial x}\left(k\frac{\partial T}{\partial x}\right) + \frac{\partial}{\partial y}\left(k\frac{\partial T}{\partial y}\right) + S(T) = 0,$$

where the temperature-dependent term represents a heat sink:

$$S(T) = -h(T - T_\infty),$$

with

$$h = 50\,\mathrm{W}/(\mathrm{m}^3\cdot\mathrm{K})$$

and

$$T_\infty = 300^\circ\mathrm{C}.$$

Boundary Conditions

The boundary conditions are:

- Left boundary:

$$T = 100^\circ\mathrm{C}$$

- Right boundary:

$$T = 500^\circ\text{C}$$

- Top boundary (insulated):

$$\frac{\partial T}{\partial y} = 0$$

- Bottom boundary (insulated):

$$\frac{\partial T}{\partial y} = 0$$

Questions

1. Write the governing equation explicitly for this problem.
2. Express the sink term in the form

$$S = S_u + S_P T_P.$$

   Identify S_u and S_P.

3. Discretize the 2D equation using the finite-volume method for a uniform grid.
4. Derive the algebraic equation for an internal control volume P in terms of

$$T_E, \quad T_W, \quad T_N, \quad T_S, \quad T_P.$$

5. Explain physically what the negative S_P means in this problem.

1 Explicit Governing Equation

Governing eqn :

$$\frac{\partial}{\partial x}(k\frac{\partial T}{\partial x}) + \frac{\partial}{\partial y}(k\frac{\partial T}{\partial y}) + S(T) = 0 \tag{1}$$

Given k is constant:

$$k(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2}) - h(T - T_\infty) = 0 \tag{2}$$

Numerical values : k=1000 ,h=50 (will be changed to 7000 for simulation to see more significant results), $T_\infty = 300$.

$$1000(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2}) - 50(T - 300) = 0 \tag{3}$$

$$(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2}) - 0.05T + 15 = 0 \tag{4}$$

2 LINEARIZED SOURCE TERM

The source term is expressed linearly as $S = S_u + S_p T_p$.
expand $S(T) = -h(T - T_\infty) = -hT + hT_\infty$
comparing with $S_u + S_p T_p$

1. $S_u = hT_\infty = 50 \cdot 300 = 15000$
2. $S_p = -h = -50$

3 Discretization using FVM

We require a uniform grid : domain divided into identical rectangular or Square cells.
Using Vertex centered FVM to define grid of nodes.
Let N_x be the number of nodes in x.
Δx = grid spacing in x = $\frac{L_x}{N_x - 1}$ = **Constant**
Let N_y be the number of nodes in y.
Δy = grid spacing in y = $\frac{L_y}{N_y - 1}$ = **Constant**
Integrate the Governing eqn (1) over a control volume $\Delta V = \Delta x \Delta y b$ centered at node P(arbitrary node)

$$\int (\frac{\partial}{\partial x}(k\frac{\partial T}{\partial x}) + \frac{\partial}{\partial y}(k\frac{\partial T}{\partial y}) + S(T))dV = 0 \tag{5}$$

Apply the divergence theorem:

$$\int \hat{n} \cdot (k\nabla T) dA + \int S(T) dV = 0 \tag{6}$$

$$kA\frac{\partial T}{\partial x}\bigg|_e - kA\frac{\partial T}{\partial x}\bigg|_w + kA\frac{\partial T}{\partial y}\bigg|_n - kA\frac{\partial T}{\partial y}\bigg|_s + (S_u + S_p T_p)\Delta V = 0 \tag{7}$$

In terms of flux the following equation can be rewritten as :

$$q_e - q_w + q_n - q_s + (S_u + S_p T_p)\Delta V = 0 \tag{8}$$

Assuming that T varies piecewise Linear between nodes we have :

1. $\frac{\partial T}{\partial x}\big|_e = \frac{T_E - T_P}{\Delta x}$
2. $\frac{\partial T}{\partial x}\big|_w = \frac{T_P - T_W}{\Delta x}$
3. $\frac{\partial T}{\partial y}\big|_n = \frac{T_N - T_P}{\Delta y}$
4. $\frac{\partial T}{\partial y}\big|_s = \frac{T_P - T_S}{\Delta y}$

Note : keep everything generalized !!

$$k_e A_e \frac{T_E - T_P}{\Delta x} - k_w A_w \frac{T_P - T_W}{\Delta x} + k_n A_n \frac{T_N - T_P}{\Delta y} - k_s A_s \frac{T_P - T_S}{\Delta y} + (S_u + S_p T_p)\Delta x \Delta y b = 0 \tag{9}$$

$$(\frac{k_e A_e}{\Delta x} + \frac{k_w A_w}{\Delta x} + \frac{k_n A_n}{\Delta y} + \frac{k_s A_s}{\Delta y} - S_p \Delta x \Delta y b) T_P = \frac{k_e A_e}{\Delta x} T_E + \frac{k_w A_w}{\Delta x} T_W + \frac{k_n A_n}{\Delta y} T_N + \frac{k_s A_s}{\Delta y} T_S + S_u \Delta x \Delta y b \tag{10}$$

4 Derive the algebraic equation for an internal control volume P

1. $A_e = \Delta y b$
2. $A_w = \Delta y b$
3. $A_n = \Delta x b$
4. $A_s = \Delta x b$
5. $k_e = k_w = k_n = k_s = k$

Coefficients of the generalized discretised eqaution are given below:

1. $a_E = \frac{k \Delta y b}{\Delta x}$

2. $a_W = \frac{k\Delta yb}{\Delta x}$
3. $a_N = \frac{k\Delta xb}{\Delta y}$
4. $a_S = \frac{k\Delta xb}{\Delta y}$
5. $b_P = S_u \Delta x \Delta yb$
6. $c_P = S_p \Delta x \Delta yb$
7. $a_P = a_E + a_W + a_N + a_S - c_P$

For interior Nodes :

$$a_P T_P = a_E T_E + a_W T_W + a_N T_N + a_S T_S + b_P \tag{11}$$

5 Physical meaning of a NEGATIVE S_p

S_p represents how strong the source reacts to a local change in temperature T_P. A negative S_p value represents a heat sink(takes away heat from the system).In this case the sink strength increases linearly with higher local temperature T_p. As T increases the Sink becomes a stronger sink means that it takes more heat away locally.
If T decreases the sink becomes a weaker sink means it takes less heat away locally.

6 Treating Top boundary

This special treatment is for the interior nodes along the top insulated edge (excluding corners).
Here the control Volume centered at a Node P is halved because the height of the CV is halved , $\frac{\Delta y}{2}$. Therefore the area of the faces is also halved.
$V = \frac{\Delta x \delta yb}{2}$, $A_e = A_w = \frac{\Delta yb}{2}$, $A_s = \frac{\Delta xb}{2}$
North face flux, $q_n = 0$ because it is insulated.
From the general flux equation we get :

$$q_e - q_w - q_s + (S_u + S_p T_p)\Delta V = 0 \tag{12}$$

With :

1. $q_e = \frac{k\Delta yb}{2\Delta x}(T_E - T_P)$
2. $q_w = \frac{k\Delta yb}{2\Delta x}(T_P - T_W)$
3. $q_s = \frac{k\Delta xb}{\Delta y}(T_P - T_S)$

$$(\frac{k\Delta yb}{\Delta x}+\frac{k\Delta xb}{\Delta y}-\frac{S_p\Delta x\Delta yb}{2})T_P=\frac{k\Delta yb}{2\Delta x}T_E+\frac{k\Delta yb}{2\Delta x}T_W+\frac{k\Delta xb}{\Delta y}T_S+\frac{S_u\Delta x\Delta yb}{2} \tag{13}$$

Coefficients of the generalized discretised eqaution are given below:

1. $a_E=\frac{k\Delta yb}{\Delta x}$
2. $a_W=\frac{k\Delta yb}{\Delta x}$
3. $a_S=\frac{k\Delta xb}{\Delta y}$
4. $b_P=S_u\Delta x\Delta yb$
5. $c_P=S_p\Delta x\Delta yb$
6. $a_P=\frac{a_E}{2}+\frac{a_W}{2}+a_S-\frac{c_P}{2}$

For the Top boundary nodes except corners the following equation holds true :

$$a_PT_P=\frac{a_E}{2}T_E+\frac{a_W}{2}T_W+a_ST_S+\frac{b_P}{2} \tag{14}$$

7 Treating bottom boundary

This special treatment is for the interior nodes along the bottom insulated edge (excluding corners). Here the control Volume centered at a Node P is halved because the height of the CV is halved , $\frac{\Delta y}{2}$. Therefore the area of the faces is also halved.
$V=\frac{\Delta x\delta yb}{2}$, $A_e=A_w=\frac{\Delta yb}{2}$, $A_n=\frac{\Delta xb}{2}$
South face flux, $q_s=0$ because it is insulated.
From the general flux equation we get :

$$q_e-q_w+q_n+(S_u+S_pT_p)\Delta V=0 \tag{15}$$

With :

1. $q_e=\frac{k\Delta yb}{2\Delta x}(T_E-T_P)$
2. $q_w=\frac{k\Delta yb}{2\Delta x}(T_P-T_W)$
3. $q_n=\frac{k\Delta xb}{\Delta y}(T_N-T_P)$

$$(\frac{k\Delta yb}{\Delta x}+\frac{k\Delta xb}{\Delta y}-\frac{S_p\Delta x\Delta yb}{2})T_P=\frac{k\Delta yb}{2\Delta x}T_E+\frac{k\Delta yb}{2\Delta x}T_W+\frac{k\Delta xb}{\Delta y}T_N+\frac{S_u\Delta x\Delta yb}{2} \tag{16}$$

Coefficients of the generalized discretised eqaution are given below:

1. $a_E=\frac{k\Delta yb}{\Delta x}$
2. $a_W=\frac{k\Delta yb}{\Delta x}$

3. $a_N = \frac{k\Delta x b}{\Delta y}$
4. $b_P = S_u \Delta x \Delta y b$
5. $c_P = S_p \Delta x \Delta y b$
6. $a_P = \frac{a_E}{2} + \frac{a_W}{2} + a_N - \frac{c_P}{2}$

The generalized discretised equation for the boundary bottom nodes are :

$$a_P T_P = \frac{a_E}{2} T_E + \frac{a_W}{2} T_W + a_N T_N + \frac{b_P}{2} \tag{17}$$

8 Results

Numerical simulations shows that Heat is transfered only in the x-direction, this is can be understood from the point of view of the boundary conditions which states that there is no heat diffusion in the y direction across the top and bottom faces of the domain. Also the right and left boundary conditions states that the tempererature is uniform in y-direction.

8.1 Simulation Results with SINK

The figure shows the temperature distribution with a sink in the system.

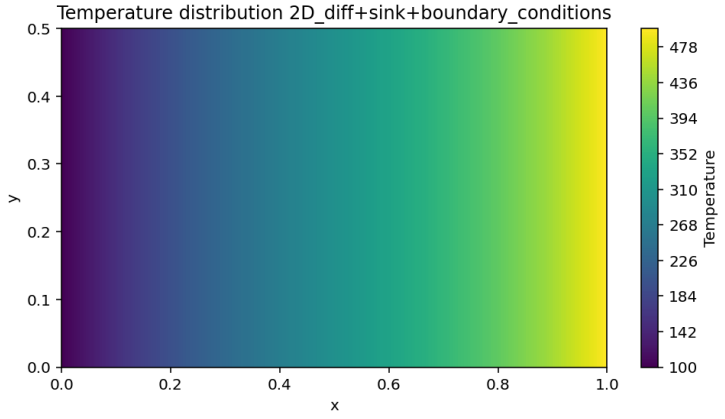


Figure 1: Temperature distribution for 2D diffusion with sink and boundary conditions

The following table shows the temperature values for the nodes in x on the 2D grid for the system **with** a sink .

Node	Distance	Numerical solution
1	0.0	100.0
2	0.111111	159.953757
3	0.222222	207.804753
4	0.333333	247.688258
5	0.444444	283.050996
6	0.555556	316.949004
7	0.666667	352.311742
8	0.777778	392.195247
9	0.888889	440.046243
10	1.0	500.0

Table 1: My data for 2D diffusion,sink ,boundary conditions

8.2 Simulation Results without SINK

The figure shows the temperature distribution without a sink in the system.

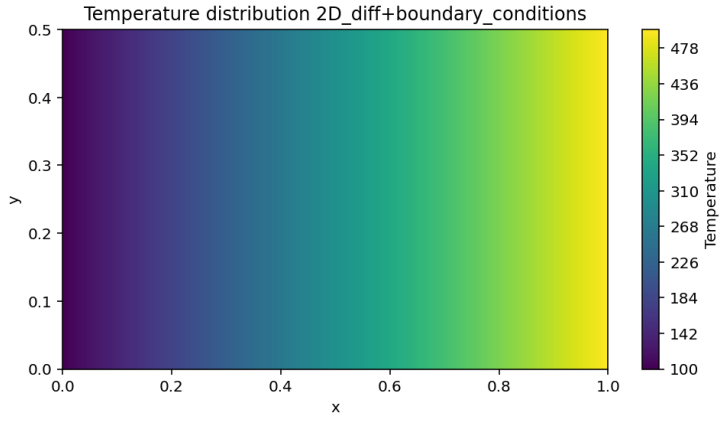


Figure 2: Temperature distribution for 2D diffusion without sink and boundary conditions

The following table shows the temperature values for the nodes in x on the 2D grid for the system **without** a sink .

Node	Distance	Numerical solution
1	0.0	100.0
2	0.111111	144.444444
3	0.222222	188.888889
4	0.333333	233.333333
5	0.444444	277.777778
6	0.555556	322.222222
7	0.666667	366.666667
8	0.777778	411.111111
9	0.888889	455.555556
10	1.0	500.0

Table 2: My data for 2D diffusion,boundary conditions

9 Discussion of results

The temperature is transported by diffusion only along x-axis, no y-variation. The SINK model match perfectly the numerical results.
Sink model:

$$S(T) = -h(T - T_\infty) \tag{18}$$

Given that $T_\infty = 300$, So

$$S(T) = -h(T - 300) \tag{19}$$

So the model predicts a linear relationship between source and temperature and this is exactly what we get.

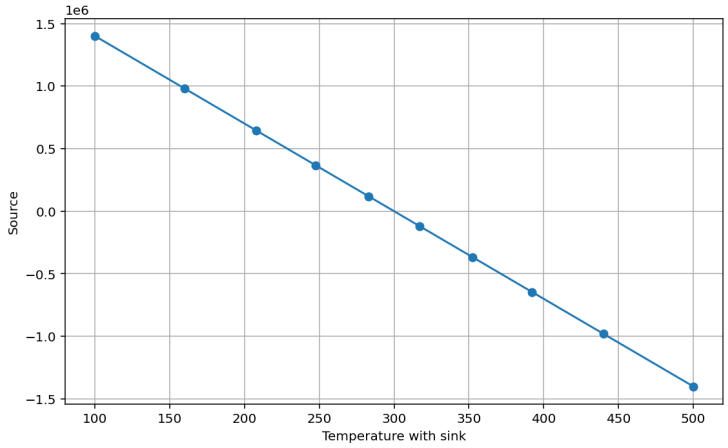


Figure 3: Source vs Temperature with sink

Furthermore :

1. for $T > 300$, S is negative implies that S acts as a **Local** Sink , takes away heat from system resulting in a decrease in temperature.

2. for $T < 300$, S is positive implies that S acts as a **Local** Source , gives heat to the system resulting in an increase in temperature.

The following table compares the case of the system with Sink and Without Sink and we investigate the local Source/Sink effect for different values of T.

Node	Temperature NO sink	Temperature With sink	Source/Sink +/-	Source
1	100.0	100.0	0.0	1400000.0
2	144.444444	159.953757	15.509312999999992	980323.701
3	188.888889	207.804753	18.915864	645366.7289999999
4	233.333333	247.688258	14.35492499999998	366182.1940000001
5	277.777778	283.050996	5.273217999999986	118643.02800000002
6	322.222222	316.949004	-5.273217999999986	-118643.02800000002
7	366.666667	352.311742	-14.354925000000037	-366182.19399999984
8	411.111111	392.195247	-18.915864	-645366.7289999999
9	455.555556	440.046243	-15.50931300000002	-980323.701
10	500.0	500.0	0.0	-1400000.0

Table 3: My data for Sink/Source Effect

The Source/Sink column shows that for temperatures(ref:Temperature NO sink) that are < 300,heat gets added to the system resulting in higher temperatures for node 1 to 5 when compared to the temperatures of the system without Sink.
A plot of Source/Sink effect and Temeperature of No-sink system is showned below:

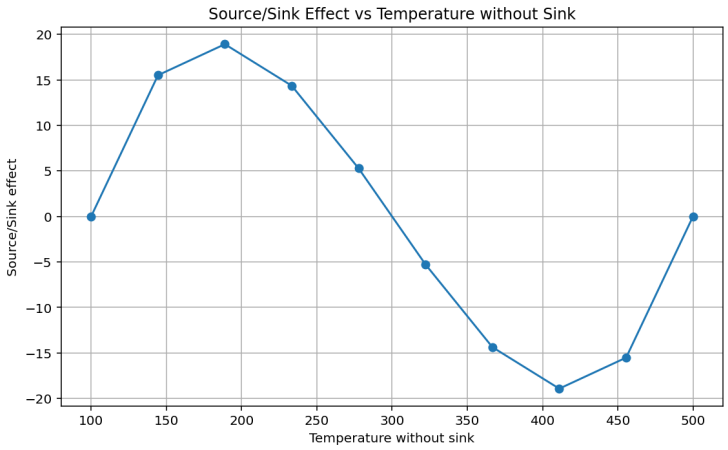


Figure 4: Source/Sink vs Temeperature of No-sink system

The Source shifts from being a Source to become a sink at T=300 which match perfectly with the predicted behaviour of the source term and exhibits a strong anti-symmetry about the critical point.